# Jason Rosenbaum Goldsmith, MD, PhD

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Highly motivated physician-scientist driven to develop novel pharmaceuticals to improve patient care. Focused on inflammatory and oncologist disorders. Extensive expertise with *in vivo* gastrointestinal models, subsequent tissue harvesting, and analyses with follow-up *in vitro* analyses. Experience working on immune-regulatory signaling to develop immune-oncology targets. Experienced in assay development. Experience with translational medicine using human samples and designing patient stratification databases. History of productive collaborations that drive innovation with 16 publications, 14 presentations at international conferences, and 8 grants. Led a medical team in a GMP environment. Experience as a technical consultant for industry collaborators.

| **Disease Models** (12 yrs) | **Biochemistry, Mol, & Cell Biology** (16 yrs) | | **Clinical Trials** (7 yrs) |
|---|---|---|---|
| IBD & ischemia<br>bone marrow chimeras<br>colitis assoc. colon cancer | flow cytometry<br>IHC & IF<br>3D cell culture | RNAseq, bulk & sc<br>enzyme kinetics<br>FRET, SPR, SPA | IRB approval<br>design & recruiting<br>sample analysis |

## EDUCATION

**MD** 2014
University of North Carolina at Chapel Hill, Chapel Hill NC

**PhD, Pharmacology** 2012
University of North Carolina at Chapel Hill, Chapel Hill NC
*Mechanisms of mu opioid receptor mediated protection from multiple models of acute injury*

**BS, Biochemistry & Chemistry**
2006
**MS, Biochemistry**
Brandeis University, Waltham MA
*Unwinding the kinetic complexity of RecBCD, a DNA helicase and nuclease found in E. coli: a single-molecule study of RecBCD$^{K177Q}$-bio*

## RESEARCH EXPERIENCE

7/2016-present **Post-doctoral Fellow**, Chen Lab, Institute for Immunology, Dept. of Pathology and Lab Medicine, University of Pennsylvania, Philadelphia PA

- Discovered the mechanism by which various TIPE family proteins regulate:
  - intestinal injury and regenerative responses
  - immune cell function and chemotaxis to inflamed tissues by modulating PI3K-Akt signaling
  - immune function which alters the disease course of colitis-associated colon cancer
- Developed multiple assays for the qualification and optimization of small molecule inhibitors for the TIPE family of proteins
- Directly supervised two undergraduate research assistants
- Mentored undergraduate, graduate students, and several junior post-doctoral fellows.
- Co-President of the Biomedical Postdoctoral Council
  - Developed and implemented the first strategic plan of the council and implemented an annual leadership development program
  - Initiated a program to provide unified teaching opportunities and didactic training for postdocs
- Established three intramural collaborations and two extramural collaborations

6/2014-6/2016 **Resident,** Dept. of Internal Medicine, Higgins & Spence Labs, University of Michigan Health Systems, Ann Arbor MI

- Directed a drug-switch clinical trial for IBD patients. Secured external funding and obtained IRB approval, helped design the clinical protocol, and wrote the final manuscript.
- Created IBD-drug response phenotype definitions , and initiated a project to develop the technology to culture enteroids from associated preserved patient biopsies for a new national IBD patient database to employ in drug response modeling and predictive biomarker studies.
- Published reviews on the immunomodulatory role of Vitamin D, and the current state of the IBD drug development pipeline.
- Led medical intern education programming
- Served on the House Officer Council to provide input to the administration regarding physician-scientist needs.

6/2006-8/2012 **Doctoral Candidate**, Jobin Lab, Department of Pharmacology, University of North Carolina at Chapel Hill, Chapel Hill NC

- Discovered that acute mu-opioid signaling is protective against multiple models of intestinal inflammation (DSS colitis and ischemia/reperfusion) and determined the mechanism of action.
- Developed a novel chemical model of larval zebrafish colitis designed to be used for screening and determined the mechanism of action by which colitis was initiated.
- Designed and executed the *in vivo* experiments and subsequent analyses to determine the mechanisms by which the natural product oxymatrine protects against murine DSS colitis.
- Determined the preliminary mechanism by which a probiotic treated IBS-D in a human clinical trial using patient samples.
- Completed a 3-year acting internship in an Advanced Outpatient IBD Clinic

## OTHER EXPERIENCE

8/2016-present **Hospitalist**
Abington Jefferson Hospital through OnCall Physician Staffing, Abington, PA

12/2016-9/2019 **Center Medical Director & Laboratory Director**
Biomat USA/ Grifols, Fairless Hills, PA

2019-present **Technical Consultant**, Gerson Lehrman Group

2016-2020 **Contract Editor**, American Journal Experts

2016 **Consultant**, QuantiaMD: case presentation development

2014 **Technical Consultant**, Protagonist Pharmaceuticals

1/2007-12/2014 **Assistant/Deputy Medical Examiner**
North Carolina Department of Health and Human Services, Raleigh NC

2007-2010 **Hospital Ethics Committee Member**
University of North Carolina at Chapel Hill, Chapel Hill NC

## LEADERSHIP, COMMITTEES & VOLUNTEERING

2016-present **Co-President,** Former Public Relations Chair and Advocacy Chair, Biomedical Postdoctoral Council, University of Pennsylvania

2015-2016 **Resident Representative,** Unified Clinical Communications Project, UMHS

2014-2016 **Physician Scientist Representative,** House Officer Council, UMHS

2014-present **"Power of Two" patient mentor**, Crohn's and Colitis Foundation of America

2010-2012 **LCME review committee**, Student steering committee; education sub-committee, University of North Carolina at Chapel Hill

2007-2008 **DNA Day Ambassador**, University of North Carolina at Chapel Hill

2006-2014 **Founder & President**, Self-Defense and MMA Club, UNC at Chapel Hill

2006-2008 **President** (2007-2008), Awareness of Domestic Violence and Sexual Assault Club

2005-2006 **Steering Committee Member,** Shapiro Science Center, Brandeis University

2004-2006 **Advisor to the Board of Trustees,** Socially responsible investing of the endowment, Brandeis University

## HONORS AND AWARDS

4/2020 American Society of Investigative Pathology Sobel Scholar Trainee Travel Award
2019-present Sigma Xi Scientific Research Honor Society
5/2019 AGA Recognition of Presentation, Early Career Investigator, Digestive Disease Week
6/2018 AGA Recognition of Presentation, Early Career Investigator, Digestive Disease Week
6/2017 Poster Award, 18th Annual Symposium, NIH Center for Molecular Studies in Digestive and Liver Diseases
8/2015 "Making a Difference." Patient Care Award, Univ. Michigan Health System
2/2013 "Real Doctors, Real People." University of North Carolina, Chapel Hill
2006-present Phi Betta Kappa Honor Society
2006 Suma Cum Laude with High Honors, Brandeis University
2006 William P. Jencks Prize in Biochemistry, Brandeis University
2006 Elihu S. Silver Prize, Brandeis University
2005 Emily Dudek Undergraduate Teaching Award, Brandeis University
2002-2006 Dean's List, Brandeis University
2002-2006 National Merit Scholar
2002-2006 Justice Brandeis Scholar (full tuition scholarship)

## PROFESSIONAL SOCIETIES

2016-present Pennsylvania Medical Society
2015-present American College of Physicians
2011-present American Gastroenterological Association
2006-present American Medical Association
2006-present American Chemical Society

## CERTIFICATION & LICENSURE

2016-present CLIA Laboratory Director
2016-present Active Medical License, Pennsylvania
2016-present Active DEA License

**INTERESTS:** Board games; 5th degree Black Belt in Kajukenbo Kung Fu (business founder/owner, head instructor); gardening

## GRANTS

2017-present NIDDK F32 DK116528 $65,128
Project: The role and mechanism of TNFAIP8 in intestinal inflammation and wound healing
**Principle Investigator**

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|---|---|---|
| 2015-2017 | BCBS of Michigan Foundation Grant<br>Project: 5-ASA to sulfasalazine drug switch: a cost savings program in ulcerative colitis patients.<br>**Principal grant writer, co-Investigator** | $70,871 |
| 2011-2014 | F30 MD/PhD Fellowship from NIDDK (5F30DK085906-02)<br>Project: The role of the mu-opioid receptor in inflammatory bowel disease pathology and therapy<br>**Principal Investigator** | $102,744 |
| 2005 | Barry S. Goldwater Scholarship<br>Federal award for college graduates pursuing a career in science, mathematics, or engineering | $7,500 |
| 2004 | IGERT Fellowship, Brandeis University | $5,000 |
| 2004 | Jerome Schiff Undergraduate Fellow, Brandeis University | $2,000 |
| 2003 | Howard Hughes Undergrad Research Fellowship, Brandeis U. | $5,000 |

## PUBLICATIONS

*Peer-Reviewed Journals and Publications*

**Jason R. Goldsmith**, Nina Spitofsky, Ali Zamani, Ryan Hood, Amanda E. Boggs, Xinyuan Li, Mingyue Li, Elizabeth Reiner, Arshad Ayyaz, Javier Rivera Guzman, Terry Cathoupolis, Hakon Hakonarson, Honghong Sun, Jeffrey L. Wrana, Michael V. Gonzalez, Youhai H. Chen. *TNFAIP8 maintains the intestinal homeostasis necessary for proper injury responses*. Nature Communications, 2020; 11:2591.

Xinyuan Li, George Luo, Ting Li, Honghong Sun, Wu Wang, Emily Eiler, **Jason R. Goldsmith**, Youhai H Chen. *The c-Rel-c-Myc axis controls metabolism and proliferation of human T leukemia cells.* Molecular Immunology, 2020; 125, 115-122.

Honghong Sun, Mei Lin, Ali Zamani, **Jason R. Goldsmith**, Amanda Boggs, Mingyue Li, Xu Chen, Xinyuan Li, Chin-Nien Lee, Brigid Dorrity, Ning Li, Yunwei Lou, Songlin Shi, Wei Wang, Youhai H. Chen. *The TIPE molecular pilot that directs lymphocyte migration in health and inflammation*. Scientific Reports. 2020; 10: 6617.

**Jason R. Goldsmith**, R Balfour Sartor. *Opioid Toxicity in Inflammatory Bowel Disease Patients Likely Includes Direct Enterocyte Effects That Exacerbate Disease.* Clinical Gastroenterology and Hepatology. 2018; 16(10): 1679-80.

**Jason R. Goldsmith**, et al. *5-ASA to Sulfasalazine Drug Switch Program in Ulcerative Colitis Patients.* American Journal of Managed Care. 2018; 24 (8 Spec No), SP303-8.

Svetlana A Faygnerts, Zhaojun Wang, Ali Zamani, Honghong Sun, Amanda E Boggs, Thomas P Porturas, Weidong Xie, Mei Lin, Terry Cathopoulis, **Jason R. Goldsmith**, Anastassios Vourekas, Youhai H. Chen. *Direction of leukocyte polarization and migration by the phosphoinostide-transfer protein TIPE2.* Nature Immunology. 2017; 18:1353-1360.

**Jason R. Goldsmith**, Youhai Chen. Review: *Regulation of Inflammation and Tumorigenesis by the TIPE Family of Phospholipid Transfer Proteins*. Cellular & Molecular Immunology. 2017; 4:1-6.

**Jason R. Goldsmith.** Review: *Vitamin D as an immunomodulatory: risks with deficiencies and benefits of supplementation.* Healthcare. 2015; 3(2): 219-232.

**Jason R. Goldsmith**, R. Balfour Sartor. Letter. *Response to the letter by Brotherton regarding "Insoluble fiber and intestinal microbiota metabolism.* Journal of Gastroenterology. 2015 Feb.

**Jason R. Goldsmith**, Tamar Ringel-Kulka, Ian Carroll, Jennica P. Siddle, Christian Jobin, Silvana P. Barros, Yehuda Ringel. *Lactobacillus Acidophilus NCFM Affects Colonic Mucosal μ Opioid Receptor Expression in Human Patients with Abdominal Pain*. Alimentary Pharmacology & Therapeutics. 2014 Jul; 40(2):200-7.

**Jason R. Goldsmith**, R. Balfour Sartor. Review: *You are what your microbiota eat: the role of diet on microbiota metabolism and the downstream impacts on host immune function and health.* Journal of Gastroenterology. 2014; 49(5): 785-798.

**Jason R. Goldsmith**, Maureen Kelly, Katherine B. Freeman, Debora Duro. *Case Report: Restitution-based therapy to induce remission in immunosuppressive refractory Crohn's Disease*. Journal of Pediatric Gastroenterology and Nutrition. 2015; 60 (4): e31-4.

Javier Rivera Guzman, JS Koo, **Jason R. Goldsmith**, Marcus Mülhbauer, Christian Jobin. *Oxymatrine prevents NF-kB nuclear translocation and ameliorates acute intestinal inflammation.* Scientific Reports. 2013; 3:1629.

**Jason R. Goldsmith**, Ernesto Perez-Chanona, Prem N. Yadav, Jennifer Whistler, Bryan Roth, Christian Jobin. *Intestinal epithelial cell-derived mu-opioid receptor signaling protects against intestinal ischemia reperfusion injury through PI3K Signaling.* AJP. 2013; 182(3):776-85.

**Jason R. Goldsmith**, Jordan L. Cocchiaro, John F Rawls, Christian Jobin. *Zebrafish glafenine-intestinal injury is ameliorated by mu-opioid signaling via enhancement of At6-dependent cellular stress responses*. Dis Models Mech. 2013; 6(1): 146-59.

**Jason R. Goldsmith**, Christian Jobin. Review: *Think small: zebrafish as a model system of human pathology.* Journal of Biotechnology and Biomedicine. 2012; 2012:817314.

**Jason R. Goldsmith**, Joshua M. Uronis, Christian Jobin. *Mu opioid signaling protects against acute murine intestinal injury in a manner involving STAT3 signaling*. AJP. 2011; 179(2): 673-83.

*Non-peer Reviewed Publications*

**Jason R. Goldsmith,** Peter D Higgins. Review: *The current state of the inflammatory bowel disease pipeline: a promising time.* Practical Gastroenterology. 2015; 92: 20-7.

*Book Chapters*

**Jason R. Goldsmith**, S Tomkovich, C Jobin. *A rapid screenable assay for compounds that protect against intestinal injury in zebrafish larvae.* Gastrointestinal Physiology and Diseases: Methods and Protocols. 2016; 281-293

*Abstracts and Presentations*

**Jason R. Goldsmith,** et al. *TNFAIP8 is a critical regulator of enterocyte differentiation and the Sca-1⁺ injury-resistant intestinal stem program.* Abstract for **Oral Presentation**. Digestive Disease Week. 5/19/2019. San Diego, CA.

**Jason R. Goldsmith,** et al. *TNFAIP8 Deficiency Results in Protection from Intestinal Injury by Increasing Basal Levels of injury-resistant, Sca-1+/fetal like stem cells.* Abstract for **Poster Presentation**. Keystone Symposium on Plasticity. 1/28/2019. Keystone, CO.

**Jason R. Goldsmith**, et al. *826-TNFAIP8 Deficiency Protects Against Intestinal Injury through Enhanced Epithelial Cell Resistance to Apoptosis via Altered Stemness.* Abstract for **Oral Presentation**. Digestive Disease Week. 6/5/2018, Washington, D.C.

**Jason R. Goldsmith**, Maureen Kelly, Katherine B. Freeman, Debora Duro. *Case Report: Restitution-based therapy to induce remission in immunosuppressive refractory Crohn's Disease*. Abstract for **Poster Presentation**. NASPGHAN. 10/10/2013, Chicago, IL.

**Jason R. Goldsmith**, Maureen Kelly, Katherine B. Freeman, Debora Duro. *Case Report: Restitution-based therapy to induce remission in immunosuppressive refractory Crohn's Disease*. Abstract for **Poster Presentation**. UNC-CH Internal Medicine Residency Research Day. 5/24/3013, Chapel Hill, NC.

**Jason R. Goldsmith**, Jordan L. Cocchiaro, John F. Rawls, Christian Jobin. *Glafenine induced ER stress and intestinal damage is reversed by the mu-opioid agonist DALDA through enhanced autophagic responses in zebrafish.* Abstract for **Oral Presentation**. Digestive Disease Week. 5/19/2012-5/22/2012, San Diego, CA.

**Jason R. Goldsmith**, Ernesto Perez-Chanona, Prem N. Yadav, Jennifer Whistler, Bryan Roth, Christian Jobin. *Intestinal epithelial cell (IEC)-derived mu-opioid signaling is protective against ischemia reperfusion (I/R) induced injury.* Abstract for **Poster of Distinction Presentation**. Digestive Disease Week. 5/19/2012-5/22/2012, San Diego, CA.

**Jason R. Goldsmith**, Ernesto Perez-Chanona, Prem Yadav, Jennifer Whistler, Bryan Roth, Christian Jobin. *Intestinal epithelial cell (IEC)-derived mu-opioid signaling is protective against two models of acute murine intestinal injury.* Abstract for **Oral Presentation**. UNC-CH Department of Pharmacology Departmental Research Retreat. 11/8/2011, Chapel Hill, NC.

**Jason R. Goldsmith**, Jordan Cocchiaro, John F. Rawls, Christian Jobin. *The NSAID Glafenine induces intestinal apoptosis and sloughing in larval zebrafish; a novel model of intestinal injury that responds to intestinal mitogens.* Abstract for **Poster Presentation**. FASEB Summer Research Conference. 8/14/2011-8/18/2011, Steamboat Springs, CO.

Javier Rivera-Guzman, Ja-Soel Koo, **Jason R. Goldsmith**, Arianne L. Theiss, Christian Jobin. *Oxymatrine blocks Nf-kB nuclear translocation and exacerbates DSS-induced colitis*. Abstract for **Poster Presentation**. FASEB Summer Research Conference. 8/14/2011-8/18/2011, Steamboat Springs, CO.

**Jason R. Goldsmith**, Jordan Cocchiaro, John F. Rawls, Christian Jobin. *The NSAID Glafenine induces intestinal apoptosis and sloughing in larval zebrafish; a novel model of intestinal injury that responds to intestinal mitogens, including the recently discovered mu-opioid receptor.* UNC-CH MD-PhD Annual Retreat. Abstract for **Oral Presentation**. 8/5/2011-8/7/2011, Wilmington, NC.

**Jason R. Goldsmith**, Jordan Cocchiaro, John F. Rawls, Christian Jobin. *The NSAID Glafenine induces intestinal apoptosis and sloughing in larval zebrafish; a novel model of intestinal injury that responds to intestinal mitogens, including the recently discovered mu-opioid receptor.* MD-PhD National Annual Meeting. Abstract for **Oral Presentation**. 7/15/2011-7/19/2011, Keystone, CO.

Javier Rivera Guzman, J.S Koo, **Jason R. Goldsmith,** Christian Jobin. *Oxymatrine blocks Nf-kB nuclear translocation and exacerbates DSS-induced colitis.* Abstract for **Poster Presentation**. Institutional Research and Academic Career Development Award (IRACDA) National Meeting. 6/2011, Houston, TX.

**Jason R. Goldsmith**, Jordan Cocchiaro, John F. Rawls, Christian Jobin. *The neuropeptide DALDA protects against NSAID-induced intestinal injury in zebrafish larvae.* Abstract for **Poster Presentation**. Center for Gastrointestinal Biology and Disease Poster Contest, Spring 2011, Chapel Hill, NC.

**Jason R. Goldsmith**, John F. Rawls, Christian Jobin. *A novel model of glafenine-induced intestinal injury in zebrafish larvae reveals a protective role for mu-opioid signaling by enhancing autophagy responses.* Abstract for **Oral Presentation**. UNC-CH Department of Pharmacology Student and Post-Doc Lecture Series. 4/15/2011, Chapel Hill, NC.

**Jason R. Goldsmith**, John F. Rawls, Christian Jobin. *The neuropeptide DALDA protects against NSAID-induced injury in zebrafish larvae*. Abstract for **Poster of Distinction Presentation**. Digestive Disease Week. 5/7/11-5/10/11, Chicago, IL.

**Jason R. Goldsmith,** Joshua M. Uronis, Greg R. Gipson, Christian Jobin. *Mu opioid signaling and STAT3 activation protects against acute murine intestinal injury in mice*. Abstract for **Poster Presentation**. Digestive Disease Week. 5/7/11-5/10/11, Chicago, IL.

Yehuda Ringel, **Jason R. Goldsmith**, Ian M. Carroll, Jennica P. Siddle, Christian Jobin, Silvana P. Barros, Tamar Ringel-Kulka. *The probiotic bacteria Lactobacillus acidophilus NCFM® (L-NCFM) and Bifidobacterium lactis Bi-07 (B-LBi07) increase expression of intestinal opioid receptors—first evidence in humans*. Abstract for **Poster Presentation**. Digestive Disease Week. 5/7/11-5/10/11, Chicago, IL.

**Jason R. Goldsmith**. *NSAID-induced enterocolitis vs Crohn's Disease*. **Oral Presentation***.* UNC-CH GI Grand Rounds. 4/5/2011, Chapel Hill, NC.

**Jason R. Goldsmith**, John F. Rawls, Christian Jobin. *Larval zebrafish intestinal injury as a model for rapid drug screening.* Abstract for **Oral Presentation**. UNC-CH MD/PhD Program Seminar Series. 3/7/2011, Chapel Hill, NC.

**Jason R. Goldsmith**, Greg R. Gipson, Christian Jobin. *Mu-opioid signaling protects against DSS-induced injury*. Abstract for **Poster Presentation**. Center for Gastrointestinal Biology and Disease Poster Contest, 5/18/2010, Chapel Hill, NC.

**Jason R. Goldsmith**, Christian Jobin. *MOR activation protects against acute intestinal injury in a manner involving STAT3 activation.* Abstract for **Oral Presentation**. Center for Gastrointestinal Biology and Disease Research Competition, Fall 2010, Chapel Hill, NC.

**Jason R. Goldsmith**, Christian Jobin. *MOR STAT3 in IBD.* Abstract for **Oral Presentation**. UNC-CH Department of Pharmacology Student and Post-Doc Lecture Series. 12/4/2009, Chapel Hill, NC.

**Jason R. Goldsmith**, Charles Elder, Richard Hammerschlag, Remy Coeytaux. *Referral patterns and patient demand for acupuncture services at Kaiser Permanente Northwest*. Abstract for **Poster Presentation**. Society for Acupuncture Research. 11/09/07-11/10/07, Baltimore, MD.

**Jason R. Goldsmith**, Andrea Nackley, Luda Diatchenko. *Polysome profiling of COMT haplotypes*. Abstract for **Oral Presentation**. UNC-CH MD/PhD Program Seminar Series. 10/29/2007, Chapel Hill, NC.